AI Playbook and Toolkit for Public Health Departments

# Public Health System Integration Patterns

ARC 340

Public health AI rarely succeeds as a stand-alone application. The value of an AI-supported workflow depends on how well it is integrated with systems of record, data exchange services, terminology services, identity systems, dashboards, case management tools, and communication channels. This module helps learners understand the integration patterns that allow AI to support public health work without creating fragmented pilots or unsafe workarounds. The module focuses on practical integration choices rather than software engineering theory. Learners examine when point-to-point interfaces, interface engines, APIs, event queues, shared services, file exchange, bulk data access, and embedded workflow tools are appropriate. The goal is to help public health teams ask better implementation questions and recognize when a proposed AI tool is not architecturally ready for operational use.

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| **Level** | intermediate/practitioner |
| **Primary track** | Technical Architecture Track |
| **Appears in tracks** | technical-architecture, operations-data-quality, program-management, governance-security |

## Learning Objectives

- Explain why system integration is central to AI readiness in public health.
- Distinguish among common integration patterns such as point-to-point interfaces, APIs, message brokers, interface engines, event queues, and bulk exchange.
- Identify the systems of record, upstream sources, downstream systems, and shared services involved in an AI-supported workflow.
- Assess whether a proposed AI tool requires read-only access, write-back access, embedded workflow integration, or manual review.
- Identify integration risks related to reliability, duplicate entry, routing errors, privacy, security, and vendor lock-in.
- Produce an integration pattern worksheet for a public health AI use case.

## Module Overview

Public health AI rarely succeeds as a stand-alone application. The value of an AI-supported workflow depends on how well it is integrated with systems of record, data exchange services, terminology services, identity systems, dashboards, case management tools, and communication channels. This module helps learners understand the integration patterns that allow AI to support public health work without creating fragmented pilots or unsafe workarounds.

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## Definitions

- Integration pattern: A reusable approach for connecting systems, services, data flows, and workflows.
- Interface engine: Software that routes, transforms, validates, and monitors messages between systems.
- Message broker: A service that allows systems to send and receive messages through a managed queue or topic.
- Bulk data exchange: The transfer of large datasets at scheduled intervals or on demand.
- Write-back: A system action that updates, creates, or changes information in another system.

## Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

## Why This Topic Matters for Public Health AI

Integration decisions determine whether AI becomes part of a reliable public health workflow or an isolated tool that adds burden. If staff must copy and paste sensitive information between systems, the AI tool may appear helpful while increasing privacy risk, documentation gaps, and error rates. If an AI service writes back to a system of record without appropriate controls, a technical integration can become an operational safety risk.

Public health agencies often operate in mixed environments that include legacy surveillance systems, modern cloud platforms, external partners, interface engines, spreadsheets, portals, and vendor-hosted tools. AI implementation requires teams to make these environments visible and decide where automation belongs. A well-integrated workflow preserves accountability by defining what data are exchanged, what actions are allowed, what is logged, and what humans must approve.

## Public Health Example

A health department may want an AI assistant to summarize incoming case reports and create follow-up tasks for investigators. A weak integration pattern would require staff to export case data, paste it into an external tool, copy the summary, and manually create a task. A stronger pattern would retrieve approved fields from the surveillance system, generate a draft summary inside the authorized workflow, and require the investigator to approve any task creation.

The technical design matters because the AI tool is no longer just generating text. It is interacting with operational systems. The agency must decide whether the tool is read-only, whether it can create draft tasks, whether it can update the case record, and how the system logs each action.

## Technical and Operational Deep Dive

Point-to-point interfaces connect two systems directly. They can be useful when the exchange is stable and narrow, but they become difficult to maintain as the number of systems increases. In public health, too many point-to-point connections can create brittle architecture, inconsistent validation, and high maintenance burden.

Interface engines and message brokers support more scalable exchange. Interface engines are common in health data exchange because they can transform messages, route them, validate required fields, and provide monitoring. Message brokers and event queues are useful when workflows are triggered by events, such as receipt of a lab result or creation of a new case. AI tools that depend on timely data may need these patterns to avoid polling, manual refreshes, or delayed processing.

APIs are increasingly important because they allow systems to request data or services in a more modular way. FHIR APIs can support structured access to clinical information, while internal APIs may expose surveillance, laboratory, or operational data to approved applications. API-based integration requires careful attention to authentication, authorization, rate limits, payload design, error handling, and audit logs.

## Risks, Failure Modes, and Guardrails

A common failure mode is integration by workaround. Staff may use manual exports, screenshots, shared drives, or copy-and-paste workflows when a tool is not properly integrated. These practices can create privacy risk, version confusion, and gaps in the official record. A guardrail is to require integration review before operational use and to document whether manual transfer of sensitive data is prohibited.

Another failure mode is uncontrolled write-back. If an AI tool can update a record, create a task, send a message, or change a status, the agency must define approval points, rollback procedures, and audit logs. The guardrail is to separate draft support from final action unless the workflow has been approved for automation.

## Application to AI-Supported Workflows

Generative AI may require integration with document repositories or systems of record so outputs can be grounded in approved content. Predictive models may require scheduled data feeds and monitoring dashboards. Agentic workflows require the strongest integration controls because they may call tools, access data, create tasks, or trigger notifications. For each AI use case, the integration pattern should match the sensitivity and operational consequence of the action.

## Reflection Questions

Which workflows currently depend on manual transfer between systems?

Which systems are the official systems of record?

Where would API access reduce burden without increasing risk?

Which actions should remain read-only?

What logs would be needed to reconstruct an AI-supported action?

## Practical Exercise

Create an integration pattern worksheet for one AI-supported workflow. Identify the systems involved, the data exchanged, the integration method, whether the AI tool is read-only or write-back, the human approval point, and the audit log requirements.

## Expected Artifact or Evidence

Integration pattern worksheet

System-of-record and source-system list

Read/write permission recommendation

Error handling and audit log requirements

## Expected Assignment

- Integration pattern worksheet
- System-of-record and source-system list
- Read/write permission recommendation
- Error handling and audit log requirements

## Knowledge Check

- 1. Which integration risk is most concerning for an AI tool that can update a case record?
- 2. What is the best reason to assign an accountability owner?
- 3. When should an AI-supported workflow be escalated for review?
- 4. Which practice best supports public accountability?
- 5. What should happen when data sources, policy, vendor configuration, or workflow use changes?

## References and Resources


- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- NIST Artificial Intelligence Risk Management Framework and NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- WHO Ethics and Governance of Artificial Intelligence for Health guidance: https://www.who.int/publications/i/item/9789240029200
- OWASP Top 10 for Large Language Model Applications: https://owasp.org/www-project-top-10-for-large-language-model-applications/

- ONC health IT, interoperability, and information blocking resources: https://www.healthit.gov/
- Agency policies for privacy, public records, cybersecurity, procurement, civil rights, accessibility, language access, and emergency communications